

peak_ewma_p2c load test (#94)

cluster-2026-10-04 · commit 5eb6195 · Darwin 25.3.0 arm64 · 18 CPUs

- CPU per RPC at the centre point: peak_ewma_p2c 28.84 μ s vs round_robin 29.66 μ s (-2.7%); LB cost above the no-LB control: 5.09 μ s vs 5.90 μ s.
- Allocation: 208 B/RPC more than round_robin (+0.6%).
- Brownout p99 during the fault: 23.0 ms vs round_robin 451.4 ms (ratio 0.05 $\times$).
- Steady heterogeneous p99: 24.3 ms vs round_robin 95.9 ms (ratio 0.25 $\times$).
- Crash-loop error rate during the fault: 0.08% vs round_robin 4.87%.
- Black hole error rate during the fault: 0.44% vs round_robin 4.81%.

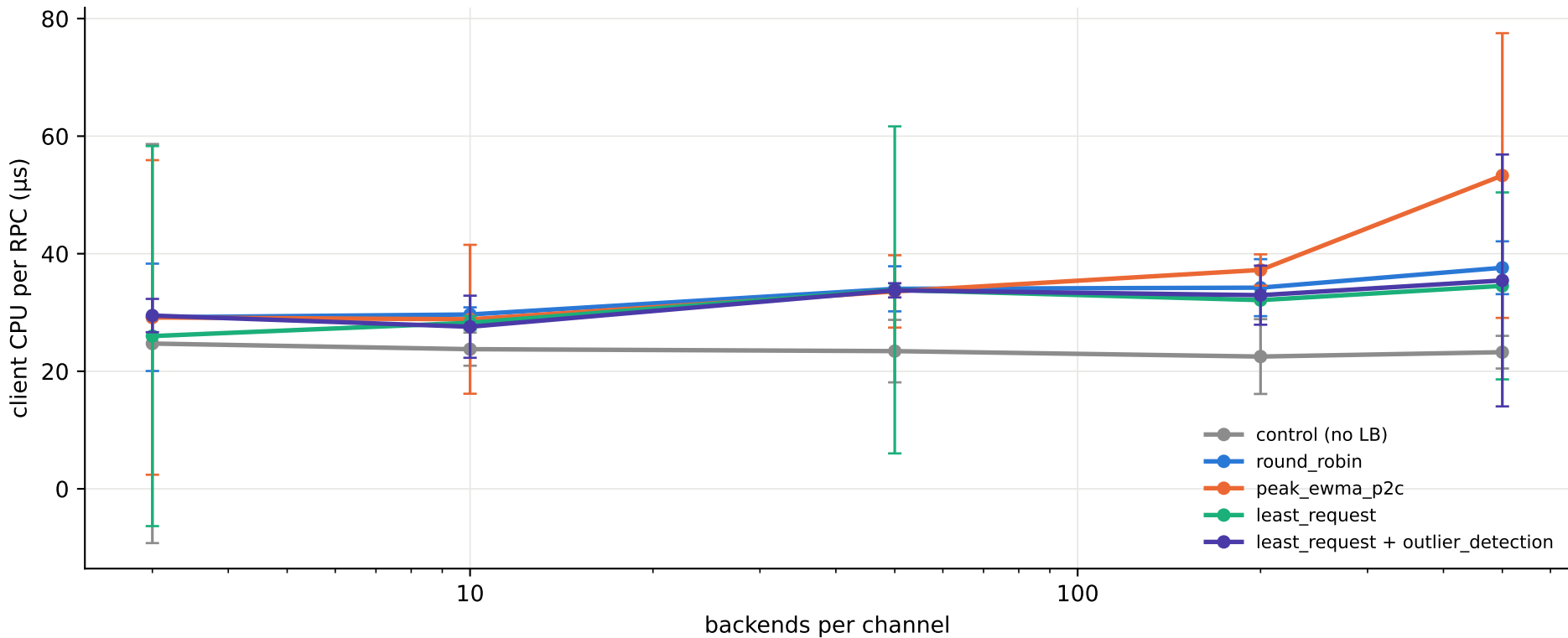
Decision table (worst-case p99 / error rate per scenario)

scenario	p99 peak (ms)	p99 least_request (ms)	p99 lr+od (ms)	p99 round_robin (ms)	errors peak	errors least_request	errors lr+od	errors round_robin	verdict (peak vs least_request)
s01_steady_het	24.3	75.4	76.3	95.9	0.25%	0.30%	0.04%	0.26%	peak: p99 68% lower than least_request
s02_steady_hom	21.6	21.5	21.6	21.5	0.00%	0.00%	0.00%	0.00%	tie
s03_brownout	23.0	25.1	22.8	451.4	0.32%	0.20%	0.10%	2.10%	lr+od: fewer errors (0.10% vs 0.32%)
s04_crashloop	21.6	21.6	21.6	21.4	0.08%	7.62%	1.76%	4.87%	peak: fewer errors (0.08% vs 1.76%);
s05_blackhole	23.1	23.1	23.0	23.0	0.44%	0.47%	0.12%	4.81%	lr+od: fewer errors (0.12% vs 0.44%)
s06_gc_pause	23.5	22.8	22.8	24.1	0.01%	0.01%	0.00%	0.01%	least_request: recovers faster (2 s vs 1 s)
s07_rolling_restart	22.5	22.4	22.6	22.4	0.00%	0.00%	0.00%	0.00%	tie
s08_scale_up_down	22.2	22.3	22.3	22.4	0.00%	0.00%	0.00%	0.00%	tie
s09_max_connections	22.1	22.1	22.0	21.9	0.00%	0.00%	0.00%	0.00%	tie
s10_node_cpu_capacity	24.1	345.2	324.3	492.2	0.71%	0.65%	0.65%	3.15%	peak: p99 93% lower than lr+od; least_request: fewer errors (0.65% vs 3.15%)
s11_client_restart	26.2	78.2	78.6	97.4	0.25%	0.29%	0.08%	0.27%	peak: p99 67% lower than least_request
s12_cross_zone	22.9	22.9	23.0	22.9	0.00%	0.00%	0.00%	0.00%	tie
s13_steady_het_1000	1030.4	1036.5	1065.1	1052.8	0.66%	1.68%	1.51%	3.97%	peak: fewer errors (0.66% vs 1.51%)

Go / no-go

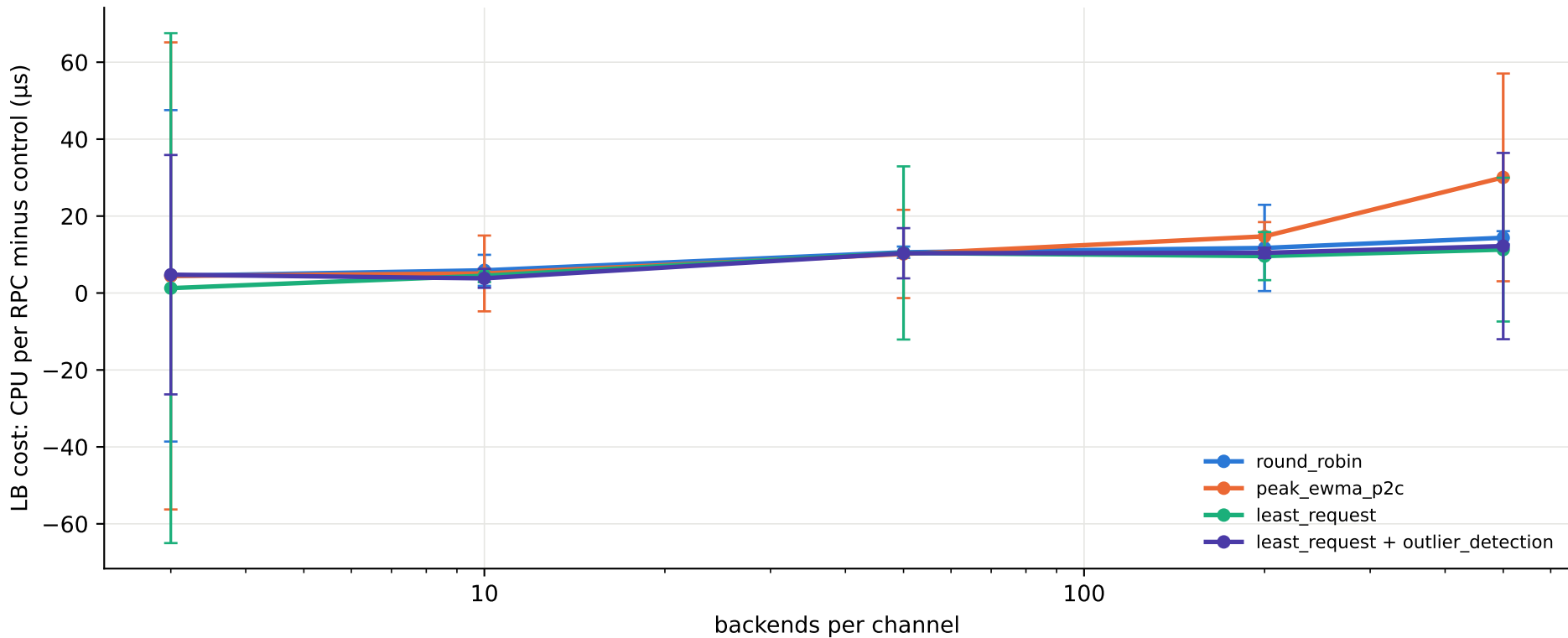
area	criterion	measured	result	note
CPU overhead	client CPU/RPC $\leq$ +25% vs ro	28.84 vs 29.66 μ s/RPC (-2.7%)	PASS	whole-process CPU incl. Netty + harness; LB co
CPU overhead	client CPU/RPC $\leq$ +25% vs ro	33.58 vs 34.03 μ s/RPC (-1.3%)	PASS	whole-process CPU incl. Netty + harness; LB co
Allocation	bytes/RPC $\leq$ +10% vs round_	32,527 vs 32,319 B/RPC (+0.	PASS	harness floor $\approx$ control's B/RPC dilutes the %; t
Memory	LB heap plateaus under 1,000	growth over last half of 240 s	FAIL	run is shorter than the 1 h in the issue
Pick time	p99 pick < 5 μ s at 500 backe	3,688 ns (sampled 1/1000)	PASS	wall time of pickSubchannel incl. the timer itse
Steady heterogeneous	p99 better than round_robin;	p99 24.3 vs 95.9 ms; errors 0	PASS	
Brownout	detected within 10 s; p99 dur	detect 3.0 s; p99 during 23.0	PASS	p99 is of successful calls; errors during the faul
Crash-loop	client error rate $\leq$ 20% of rou	0.081% vs 4.874% (0.02 $\times$)	PASS	
Black hole	client error rate $\leq$ 20% of rou	0.439% vs 4.813% (0.09 $\times$)	PASS	
Homogeneous steady	0 false ejections/hour; every	0 ejections in 6.0 client-minu	PASS	
Herd (16 clients)	max healthy-backend share $\leq$	1.33 $\times$ (p95 of 1 s windows; n	PASS	
vs least_request + outlier_de	document where each wins	s01: p99 peak (24.3/76.3 ms)	INFO	

Client CPU per RPC vs backends per channel



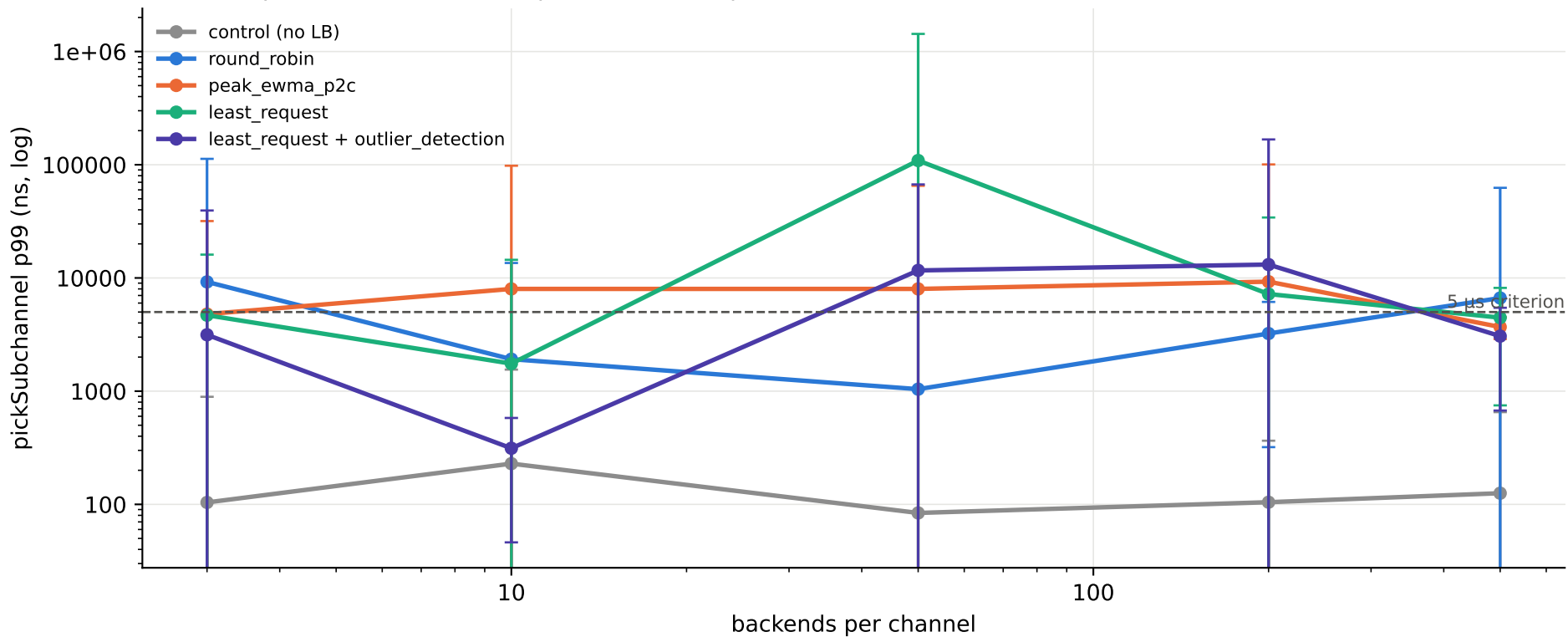
At 500 backends per channel, peak_ewma_p2c is 53.30 vs round_robin 37.61 (+42%).

LB cost (policy - control) vs backends



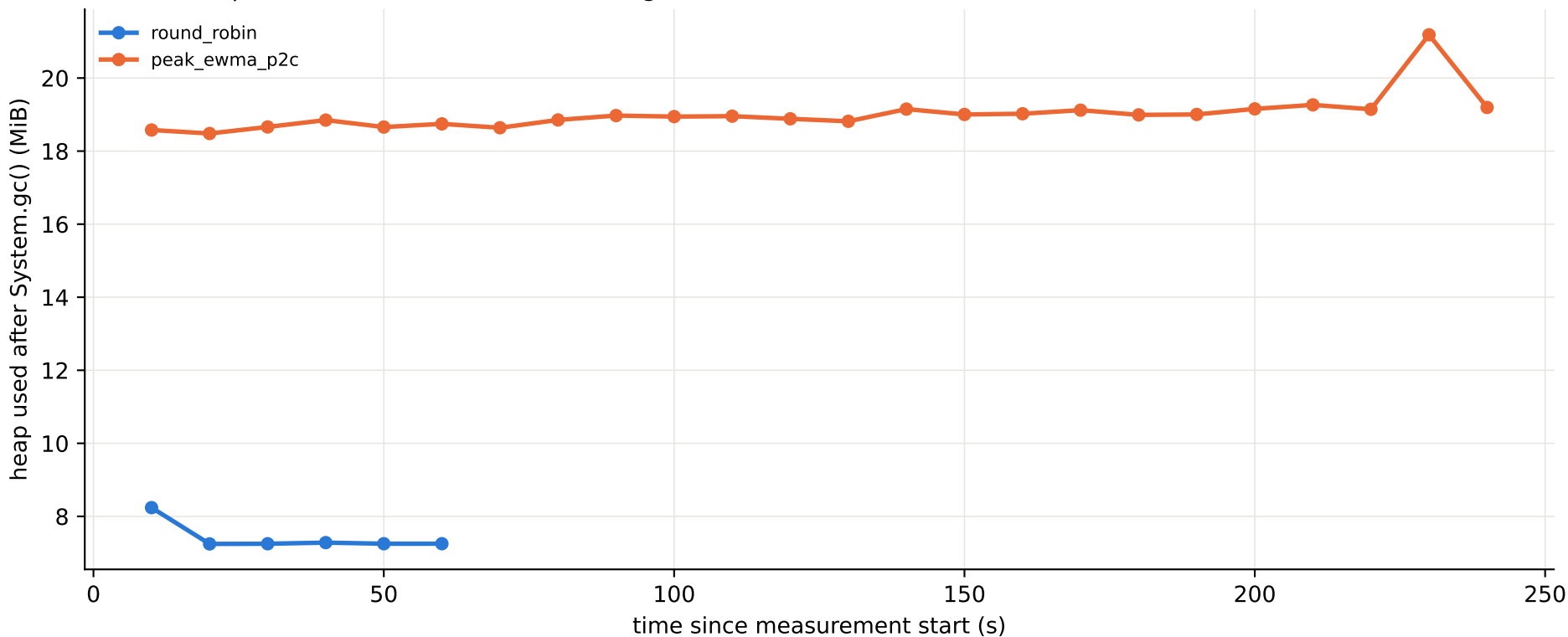
At 500 backends per channel, peak_ewma_p2c is 30.05 vs round_robin 14.37 (+109%).

Pick time p99 vs backends (sampled 1 in 1,000 picks)



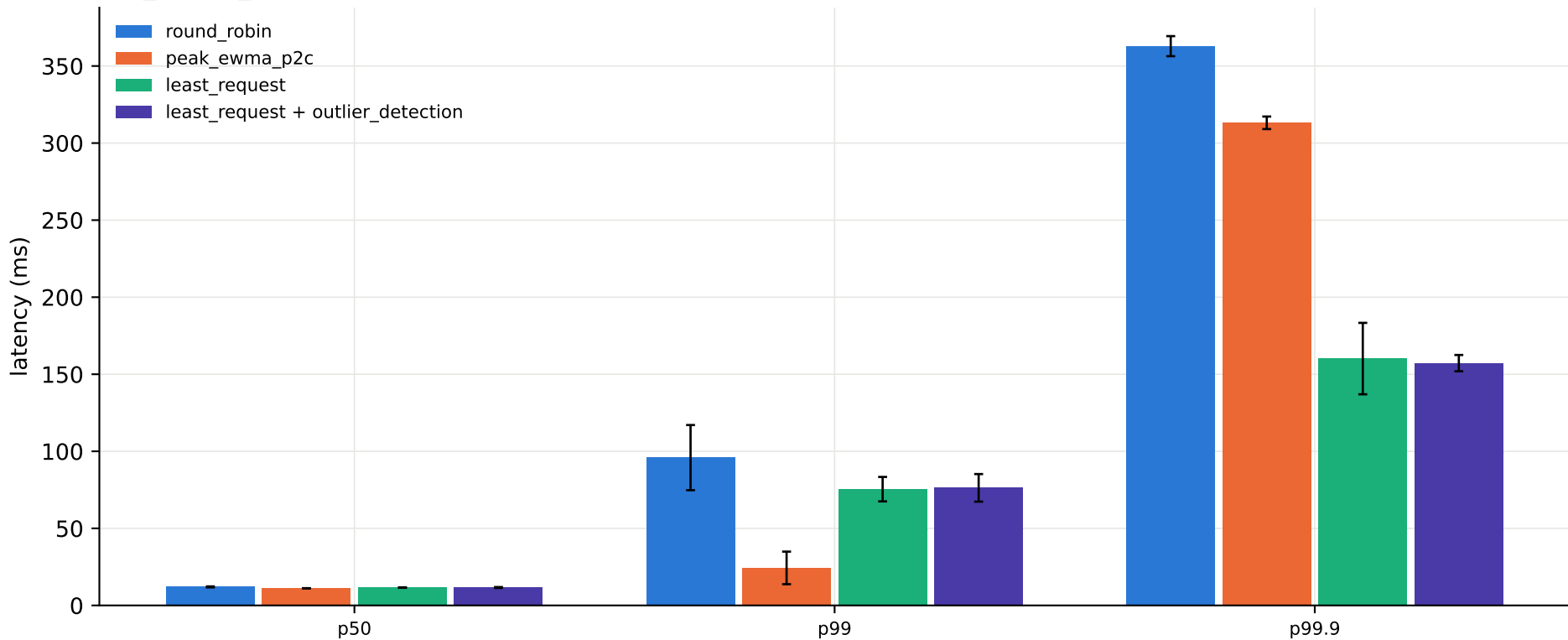
At 500 backends per channel, peak_ewma_p2c is 3688.00 vs round_robin 6649.00 (-45%).

Process heap after GC vs time, 1,000 churning methods (10 new names/s)



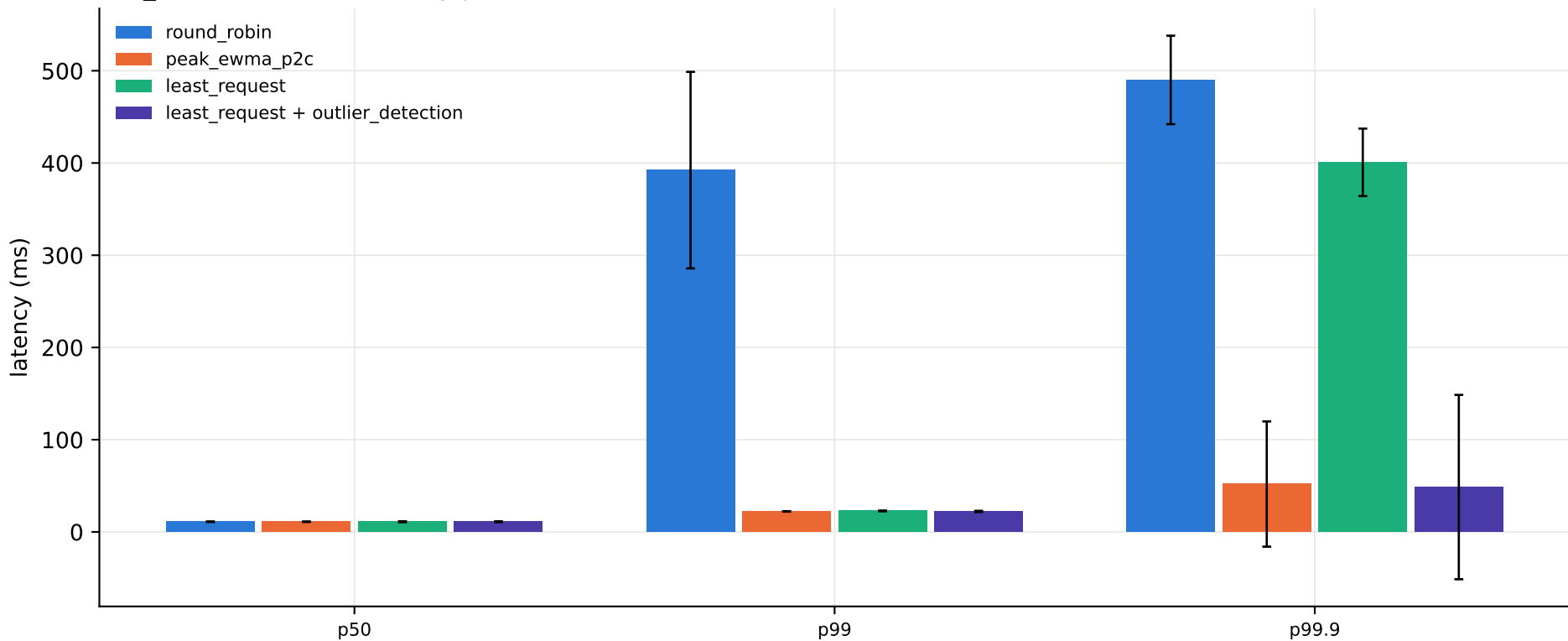
peak_ewma_p2c heap after GC goes 18.6 → 19.2 MB over 240 s; growth over the last half is +5.0% (still growing).

s01_steady_het: client latency percentiles (whole window)



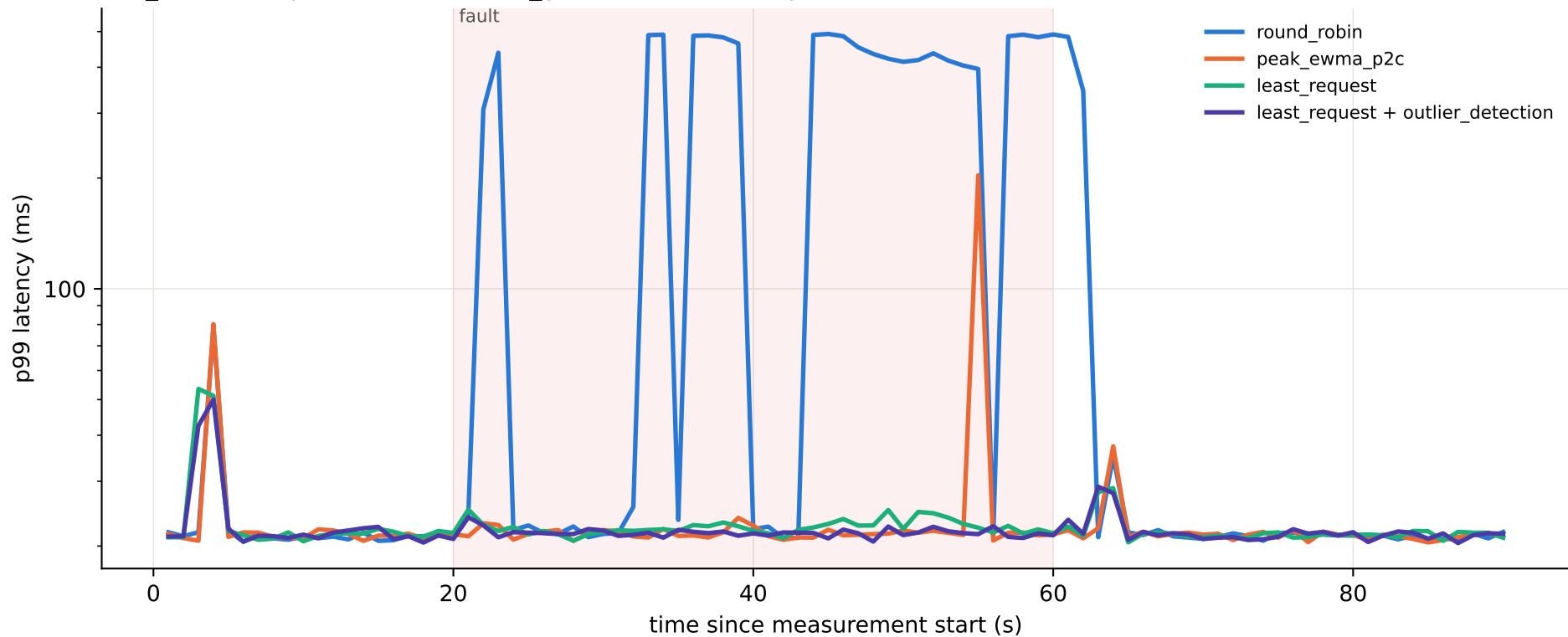
p99: peak_ewma_p2c 24.3 ms vs round_robin 95.9 ms (-75%); lowest p99: peak_ewma_p2c.

s03_brownout: client latency percentiles (whole window)



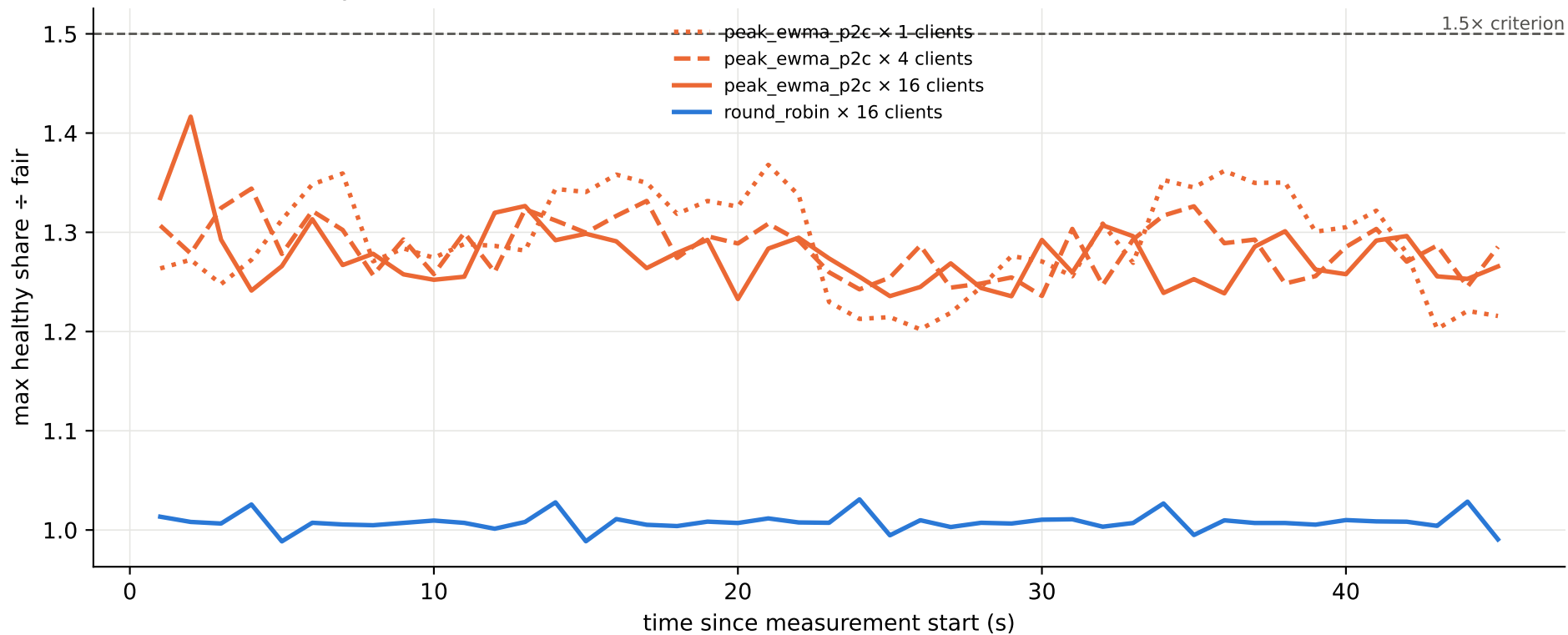
p99: peak_ewma_p2c 22.3 ms vs round_robin 392.3 ms (-94%); lowest p99: least_request + outlier_detection.

s03_brownout: p99 over time (U10_get, 1 s windows, repeat 0)



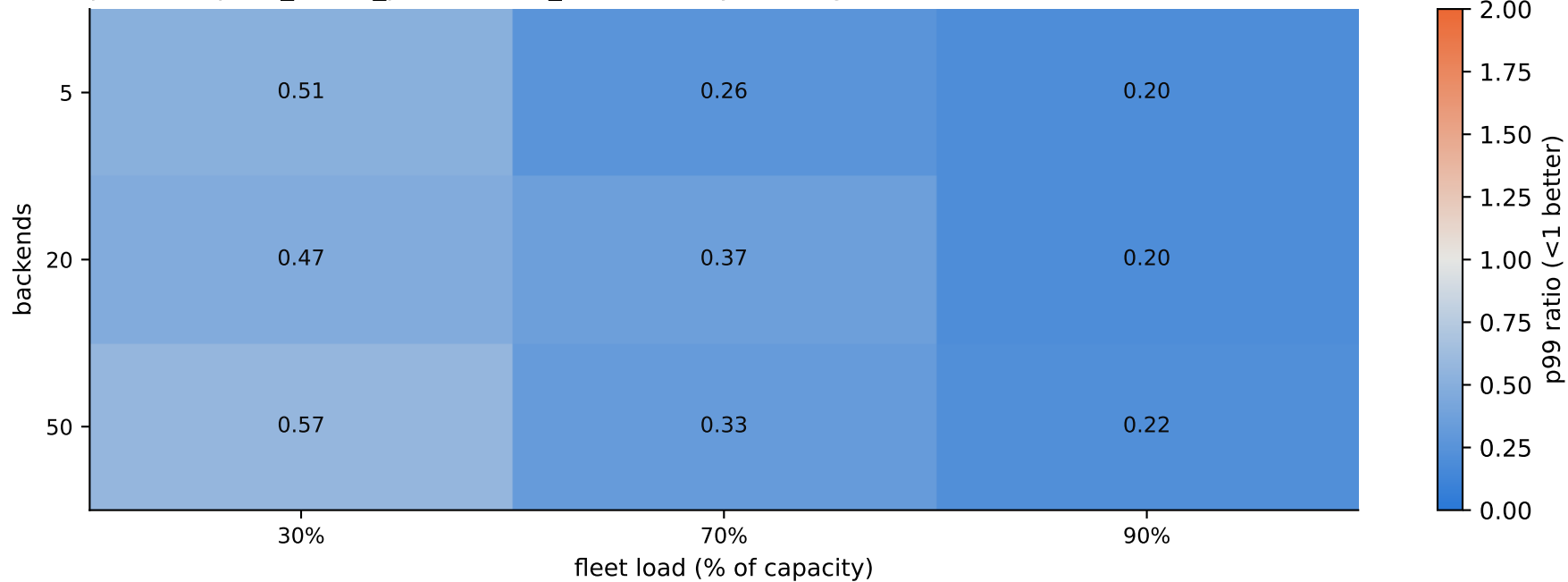
Worst 1 s p99: peak_ewma_p2c 203.5 ms, round_robin 492.7 ms.

Herd: busiest healthy backend's combined share ÷ fair share, 1 s windows



With 16 peak_ewma_p2c clients the busiest healthy backend gets 1.33x its fair share (p95 over windows).

p99 ratio peak_ewma_p2c ÷ round_robin (steady heterogeneous fleet)



peak_ewma_p2c p99 ranges from 0.20× to 0.57× round_robin's across the grid.